23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY

PHASE – I (Analysis) Problem Identification
& Class Mapping (CO1)

Group No : C7

Team Members :

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1. Problem Title : Online Course Registration System.

2. Problem Description :

Our project, Online Course Registration System allows students to choose available courses and register for them through an online platform. The project manages student details, information about courses and enrollment records. It also ensures that the students can only register for courses which has availability of seats and keeps track of registered courses. The Administrators can add or update the course details, and the students can register or drop the courses easily.

Class Name	Description
User	Base class representing a system user with login credentials
Student	Represents a student who registers for courses
Course	Represents a course available for registration
Registration	Handles the process of student enrolling in courses
Admin	Manages courses and system information
Instructor	Represents a faculty member assigned to teach courses
CourseCatalog	Maintains and manages the list of all available courses

3. Identified Entities (Classes)

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
User	username, password	login()
Student	studentID, name, email	registerCourse(), dropCourse(), viewCourses()
Course	courseID, courseName, credits, seatsAvailable	checkAvailability() , reserveSeats(), updateCourse(), viewCourseDetails()
Registration	registrationID, studentID, courseID, date, status	register(), cancelRegistration()
Admin	adminID, name	addCourse(), removeCourse(), updateCourse(), assignInstructor()
Instructor	instructorID, name	viewAssignedCourses()
CourseCatalog	courseList	displayCourses(), searchCourse(), addCourse(), removeCourse(courseID : String)